

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2025****Course: F. Y. B. Tech****Semester: I****Branch : All Branches****Subject Code & Name: Engineering Chemistry (BTBS202)****Max Marks: 60****Date:12/01/2026****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clear.

		(CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	_____ does not form good lather with soap. a. Distilled water b. Hard water c. Soft water d. Mineral water	03	1
2	For regeneration of Anion Exchange Resin _____ solution is used. a. HCl b. NaOH c. KCl d. NaCl	03	1
3	The temporary hardness of water can be removed by _____. a. Chlorination b. Calcination c. Cooling d. Boiling	03	1
4	If only one Phase is present in one component system then the degree of freedom is _____. a. 0 b. 1 c. 2 d. 3	04	1
5	_____ is the example of two component system. a. N ₂ -S ₂ b. Pb-Ag c. H ₂ O-H ₂ O(g) d. none of these	04	1
6	In water system, the curve which represents equilibrium between Solid & Vapors is called _____. a. vaporization curve b. sublimation curve c. melting point curve d. triple point	04	1
7	Corrosion is more in _____ medium. a. amphoteric b. neutral c. basic d. acidic	02	1
8	_____ corrosion occurs due to the formation of pinholes. a. Oxidation b. Microbial c. Chemical d. Pitting	02	1
9	_____ is the purest form of coal.	05	1

	a. Peat	b. Lignite	c. Anthracite	d. Coke		
10	The main constituent of fuel is _____.				05	1
	a. Carbon	b. Nitrogen	c. Sulphur	d. Oxygen		
11	Methyl orange is a weak _____ base.				01	1
	a. inorganic	b. physical	c. analytical	d. organic		
12	The byproduct of H ₂ -O ₂ fuel cell is _____.				01	1
	a. CO ₂	b. water	c. methane	d. LPG		
Q. 2	Solve the following.					12
A)	Explain the Hot Lime - Soda process of softening of water.				03	6
B)	Discuss in brief determination of Hardness by EDTA method.				03	6
Q.3	Solve the following.					12
A)	Explain Phase, component and degree of freedom with examples in Phase Rule equation.				04	6
B)	Explain in detail one component phase diagram of Water system.				04	6
Q. 4	Solve Any Two of the following.					12
A)	Explain in detail oxidation corrosion.				02	6
B)	Discuss Hydrogen evolution mechanism of corrosion.				02	6
C)	Write a note on Sacrificial Anodic Protection.				02	6
Q.5	Solve Any Two of the following.					12
A)	Discuss briefly characteristic of good fuel.				05	6
B)	Explain in detail types of lubricants.				05	6
C)	Write a note on refining of petroleum.				05	6
Q. 6	Solve Any Two of the following.					12
A)	Explain the Nernst Equation and its application for calculation for half-cell potential.				01	6
B)	Write a note on Hydrogen – Oxygen (H ₂ -O ₂) Fuel Cell.				01	6
C)	Discuss the conductometric titration with any two examples.				01	6
*** End ***						